PROGRAMS

1. Build a responsive web application for shopping cart with registration, login, catalog and cart pages using CSS3 features, flex and grid.

Responsive Web Design

Responsive web design is about creating web sites which automatically adjust themselves to look good on all devices, from small phones to large desktops. This is achieved using techniques like flexible grids, images, and CSS media queries to adjust content seamlessly. It avoids the need for separate mobile apps by ensuring usability and visual appeal across all viewports with one codebase.

.

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
<link rel="stylesheet" href="./style.css">
<title>Home - JB Online Book Store</title>
</head>
<body>
<div class="wrapper">
<div class="container">
<header>
<table width="100%" align="center" cellpadding="0" cellspacing="2">
<tr>
<th width="20%"><img src="jblogo.png" alt="JB Online Book Store LOGO" width=""
height="100"/></th>
<th colspan=4>
<h1 style="color:white;">JB Online Book Store - Best Online Ebooks Store </h1>
</th>
</tr>
</table>
</header>
<nav>
<table width="100%" align="center" cellpadding="0" cellspacing="2">
<tbody align="center" style="font-weight:bold;font-size:18px;"">
<tr>
<td width="20%"><hr><a href="index.html">Home</a><hr></td>
<td width="20%"><hr><a href="login.html">Login</a><hr></td>
<td width="20%"><hr><a href="registration.html">Registration</a><hr></td>
<td width="20%"><hr><a href="catalog.html" >Catalog</a><hr></td>
<td width="20%"><hr><a href="cart.html" >Cart</a><hr></td>
</tr>
</tbody>
</table>
</nav>
</div>
<div class="container1">
<div class="sidebar1"></div>
<div class="container2">
<main>
<center>
<h2>Welcome to JB Online Book Store</h2>
```

```
<p> Discover the  <font size=5> joy of smart shopping at </font> JB Online Book Store! <font
size=5> get ready </font>
for <font size=5>this </font>.</br>special discount season <font
size=5>save </font>  up to <font size=5>90%</font> on all your purchases. Happy shopping!
        </br></p>
<br/><br/><br/><br/><br/>
<marquee behavior="alternate" scrolldelay="fast"  direction="right"> <img src="a.jpg"
    width="100" height="130" > </marquee>
</main>
</div>
<div class="sidebar2"></div>
</div>
<footer><font color="white">(C) 2026 All rights reserved by JB Online Book Store
    </font></footer>
</div>
</body>
</html>
```

login.html

```
<!DOCTYPE html>
<html lang="en">
<head>
<link rel="stylesheet" href="./style.css">
<title>Login - JB Online Book Store</title>
</head>
<body>
<div class="wrapper">
<div class="container">
<header>
<table width="100%" align="center" cellpadding="0" cellspacing="2">
<tr>
<th width="20%"><img src="JBlogo.png" alt="JB Online Book Store LOGO" width="130"
height="100"/></th>
<th colspan=4>
<h1 style="color:white;">JB Online Book Store - Best Online Ebooks Store </h1>
</th>
</tr>
</table>
</header>
<nav>
<table width="100%" align="center" cellpadding="0" cellspacing="2">
<tbody align="center" style="font-weight:bold;font-size:18px;"">
<tr>
<td width="20%"><hr><a href="index.html">Home</a><hr></td>
<td width="20%"><hr><a href="login.html">Login</a><hr></td>
<td width="20%"><hr><a href="registration.html">Registration</a><hr></td>
<td width="20%"><hr><a href="catalog.html" >Catalog</a><hr></td>
<td width="20%"><hr><a href="cart.html" >Cart</a><hr></td>
</tr>
</tbody>
</table>
</nav>
```

```
</div>
<div class="container1">
<div class="sidebar1"></div>
<div class="container2">
<main>
<center><br>
<h3> Login Details</h3> <br/>
<form name="f1">
<table width="100%" align="center" >
<tr>
<td> User Name : </td>
<td> <input type="text" name="username"></td>
</tr>
<tr><td><br></td></tr>
<tr>
<td> Password : </td>
<td> <input type="password" name="password"></td>
</tr>
<tr><td><br></td></tr>
<tr><td></td>
<td><input type="submit" value="SUBMIT">
<input type="reset" value="RESET"></td>
</tr>
</table>
</form>
</center>
</main>
</div>
<div class="sidebar2"></div>
</div>
<footer><font color="white">(C) 2026 All rights reserved by JB Online Book Store
</font></footer>
</div>
</body>
</html>
```

registration.html

```
<!DOCTYPE html>
<html lang="en">
<head>
<link rel="stylesheet" href="./style.css">
<title>Registration - JB Online Book Store</title>
</head>
<body>
<div class="wrapper">
<div class="container">
<header>
<table width="100%" align="center" cellpadding="0" cellspacing="2">
<tr>
<th width="20%"><img src="JBlogo.png" alt="JB Online Book Store LOGO" width="130"
height="100"/></th>
<th colspan=4>
```

```
<h1 style="color:white;">JB Online Book Store - Best Online Ebooks Store</h1>
</th>
</tr>
</table>
</header>
<nav>
<table width="100%" align="center" cellpadding="0" cellspacing="2">
<tbody align="center" style="font-weight:bold;font-size:18px;"">
<tr>
<td width="20%"><hr><a href="index.html">Home</a><hr></td>
<td width="20%"><hr><a href="login.html">Login</a><hr></td>
<td width="20%"><hr><a href= "registration.html"> Registration</a><hr></td>
<td width="20%"><hr><a href="catalog.html" >Catalog</a><hr></td>
<td width="20%"><hr><a href="cart.html" >Cart</a><hr></td>
</tr>
</tbody>
</table>
</nav>
</div>
<div class="container1">
<div class="sidebar1"></div>
<div class="container2">
<main>
<center><br>
<h3>Registration Form </h3>
<br/>
<form name="f1">
<table cellpadding="1" align="center" >
<tr><td> Name:*</td>
<td><input type="text" name="username"></td></tr>
<tr><td>Password:*</td>
<td><input type="password" name="password"></td></tr>
<tr><td>Email ID:*</td>
<td><input type="text" name="email"></td></tr>
<tr><td>Phone Number:*</td>
<td><input type="text" name="phno"></td></tr>
<tr><td valign="top">Gender:*</td>
<td><input type="radio" name="radio" value="1">Male   
<input type="radio" name="radio" value="2">Female</td></tr>
<tr> <td valign="top">Language Known:*</td>
<td> <input type="checkbox" name="checkbox"
value="English">English<br/>
<input type="checkbox" name="checkbox" value="Telugu">Telugu<br>
<input type="checkbox" name="checkbox" value="Hindi">Hindi<br>
<input type="checkbox" name="checkbox" value="Tamil">Tamil
</td></tr>
<tr> <td valign="top">Address:*</td>
<td><textarea name="address"></textarea></td>
<tr><td></td><td><input type="submit" value="submit"
hspace="10">
<input type="reset" value="reset"></td></tr>
<tr> <td colspan=2 >*<font color="#FF0000">fields are
mandatory</font>
```

```
</td>
</tr>
</table>
</form>
</center>
</main>
</div>
<div class="sidebar2"></div>
</div>
<footer><font color="white">(C) 2026 All rights reserved by JB Online Book Store
 </font></footer>
</div>
</body>
</html>
```

cart.html

```
<!DOCTYPE html>
<html lang="en">
<head>
<link rel="stylesheet" href="./style.css">
<title>Cart - JB Online Book Store</title>
</head>
<body>
<div class="wrapper">
<div class="container">
<header>
<table width="100%" align="center" cellpadding="0" cellspacing="2">
<tr>
<th width="20%"><img src="JBlogo.png" alt="JB Online Book Store LOGO" width="130"
height="100"/></th>
<th colspan=4>
<h1 style="color:white;">JB Online Book Store - Best Online Ebooks Store</h1>
</th>
</tr>
</table>
</header>
<nav>
<table width="100%" align="center" cellpadding="0" cellspacing="2">
<tbody align="center" style="font-weight:bold;font-size:18px;"">
<tr>
<td width="20%"><hr><a href="index.html">Home</a><hr></td>
<td width="20%"><hr><a href="login.html">Login</a><hr></td>
<td width="20%"><hr><a href="registration.html">Registration</a><hr></td>
<td width="20%"><hr><a href="catalog.html" >Catalog</a><hr></td>
<td width="20%"><hr><a href="cart.html" >Cart</a><hr></td>
</tr>
</tbody>
</table>
</nav>
</div>
<div class="container1">
<div class="sidebar1"></div>
```

```
<div class="container2">
<main>
<center>
<h3>Cart</h3>
<table width="100%" align="center" >
<tbody>
<tr>
<th width="40%"><hr>BookName<hr></th>
<th width="20%"><hr>Price<hr></th>
<th width="20%"><hr>Quantity<hr></th>
<th width="20%"><hr>Amount<hr></th> </tr>
</tbody>
<tbody align=center>
<tr> <td>Java Programming </td>
<td>Rs. 2300/-</td>
<td>2</td>
<td>Rs. 4600/-</td></tr>
<tr><td>Web Technologies</td>
<td>Rs. 3000/-</td>
<td>1</td>
<td>Rs. 3000/-</td></tr>
<tr><td></td>
<td><hr><font color="#996600">Total Amount:</font><hr></td>
<td><hr>3<hr></td>
<td><hr>Rs. 7600/-<hr></td> </tr>
</tbody>
</table>
</center>
</main>
</div>
<div class="sidebar2"></div>
</div>
<footer><font color="white">(C) 2026 All rights reserved by JB Online Book Store
</font></footer>
</div>
</body>
</html>
```

Catalog.html

```
<!DOCTYPE html>
<html>
<head>
<script src="script.js"> </script>
<link rel="stylesheet" href="./style.css">
<title>Catalog - JB Online Book Store</title>
</head>

<body>
<div class="wrapper">
<div class="container">
<header>
<table width="100%" align="center" cellpadding="0" cellspacing="2">
<tr>
```

```
<th width="20%"><img src="JBlogo.png" alt="JB Online Book Store LOGO" width="130"
height="100"/></th>
<th colspan=4>
<h1 style="color:white;">JB Online Book Store - Best Online Ebooks Store </h1>
</th>
</tr>
</table>
</header>
<nav>
<table width="100%" align="center" cellpadding="0" cellspacing="2">
<tbody align="center" style="font-weight:bold;font-size:18px;"">
<tr>
<td width="20%"><hr><a href="index.html">Home</a><hr></td>
<td width="20%"><hr><a href="login.html">Login</a><hr></td>
<td width="20%"><hr><a href="registration.html">Registration</a><hr></td>
<td width="20%"><hr><a href="catalog.html" >Catalog</a><hr></td>
<td width="20%"><hr><a href="cart.html" >Cart</a><hr></td>
</tr>
</tbody>
</table>
</nav>
</div>
<div class="container1">
<div class="sidebar1"></div>
<div class="container2">
<main>

<center>
    <div class="catalog">
      <h2>Books Catalog</h2>
    <div class="product">
      <h3>Java Programming</h3>
      <p>₹2300</p>
      <button type="button" onclick="addToCart('Java Programming', '₹2300')">
        Add to Cart
      </button>
    </div>

    <div class="product">
      <h3> Web Technologies</h3>
      <p>₹3000</p>
      <button type="button" onclick="addToCart('Web Technologies', '₹3000')">
        Add to Cart
      </button>
    </div>
</main>
</div>
<div class="sidebar2"></div>
</div>
<footer><font color="white">(C) 2026 All rights reserved by JB Online Book Store
</font></footer>
</div>
</body>
</html>
```

```
    </div>

    <br>

    <br>
</body>
</html>
```

style.css

```
body{
  font-family: monospace;
}

main {
  background-color: #efefef;
  color: #330000;
  margin-left: 10px;
  height: 60vh;
}

 header, footer {
  background-color: #000d57;
  color: #fff;
  padding: 1rem;
  height: 50px;
}

header, nav{
  margin-bottom: 10px;
  flex-basis: 50%;
}

footer{
  margin-top: 10px;
}
nav {
  background-color: #fff;
  color: #000;
  padding: 1rem;
  height: 20px;
}

.sidebar1, .sidebar2 {
  flex-basis: 10%;
  background-color: #fff;
  color: #000;
}

.sidebar2{
  margin-left: 10px;
}

.container1{
```

```
  display: flex;
}

.container2 {
  display: flex;
  flex-direction: column;
  flex: 1;
}

header, nav, main, .sidebar1, .sidebar2, footer{
  display: flex;
  align-items: center;
  justify-content: center;
  border-radius: 10px;
}

.wrapper {
  display: flex;
  flex-direction: column;
  font-weight: 600;
}
```

Output Screens:

JB Online Book Store - Best Online Ebooks Store
Home
Login
Registration
Catalog
Cart
Login Details
User Name :
Password :
SUBMIT
RESET
(C) 2026 All rights reserved by JB Online Book Store

JB Online Book Store - Best Online Ebooks Store
Home
Login
Registration
Catalog
Cart
Registration Form
Name:*
Password:*
Email ID:*
Phone Number:*
Gender:*
Male
Female
Language Known:*
English
Telugu
Hindi
Tamil
Address:*
submit
reset
*Fields are mandatory
(C) 2026 All rights reserved by JB Online Book Store

JB Online Book Store - Best Online Ebooks Store
Home
Login
Registration
Catalog
Cart
Books Catalog
Java Programming
₹2300
Add to Cart
Web Technologies
₹3000
Add to Cart
(C) 2026 All rights reserved by JB Online Book Store

JB Online Book Store - Best Online Ebooks Store
Home
Login
Registration
Catalog
Cart
Cart
BookName | Price | Quantity | Amount
Java Programming | Rs. 2300/- | 2 | Rs. 4600/-
Web Technologies | Rs. 3000/- | 1 | Rs. 3000/-
Total Amount: | 3 | Rs. 7600/-
(C) 2024 All rights reserved by JB Online Book Store

2. **Make the above web application responsive web application using Bootstrap framework**

Registration.html

```
<!DOCTYPE html>
<html lang="en">
<head>
   <link rel="stylesheet" href="./style.css">
   <title>Registration - JBS</title>
  <script language="javascript">
   function validate() {
      // username validation
      var uname = f1.username.value;
      if (uname.length<=0)
      {
         alert("Please Enter UserName");
         f1.username.focus();
         return false;
      }
      if (uname.length < 8)
      {
         alert("Please enter UserName not less than 8");
         f1.username.focus();
         return false;
      }
      //password validation
      var pwd = f1.password.value;
      if (pwd.length<=0)
      {
         alert("Please Enter password");
         f1.password.focus();
         return false;
      }
      if (pwd.length < 6)
      {
         alert("Please enter Password not less than 6");
         f1.password.focus();
         return false;
      }
      // email validation
      var email = f1.email.value;
      if (email.length<=0)
      {
         alert("Please Enter email");
         f1.email.focus();
         return false;
      }
      else {
         let eflag=false;
         for(i=0;i<email.length;i++) {
            if(email.charAt(i)=="@")
               {
                  eflag=true;
```

```
            }
        }
        if(!(eflag))
        {
            alert("Please enter a valid Email ID");
            f1.email.focus();
            return false;
        }
    }
    // phone number validation
    var phno = f1.phno.value;
    if (phno.length<=0)
    {
        alert("Please Enter Phone Number");
        f1.phno.focus();
        return false;
    }
    if (isNaN(phno))
    {
        alert("Please Enter Valid Phone Number");
        f1.phno.focus();
        return false;
    }
    if (phno.length != 10)
    {
        alert("Please Enter Valid Phone Number");
        f1.phno.focus();
        return false;
    }
    // gender validation
    let flag=false;
    for(i=0;i<f1.gen.length;i++)
        if(f1.gen[i].checked)
            flag=true;
    if(!(flag))
    {
    alert("Please choose a Gender");
    return false;
    }
    // Language validation
    flag=false;
    for(i=0;i<f1.lang.length;i++)
        if(f1.lang[i].checked)
            flag=true;
    if(!(flag))
    {
    alert("Please select at least one of the Language options.");
    return false;
    }
    // address validation
    var addr = f1.address.value;
    if (addr.length<=0)
    {
        alert("Please Enter address");
```

```
            f1.address.focus();
            return false;
        }
        // to display Success message
        alert("Registration Successful");
    }
    </script>

</head>
<body>
    <div class="wrapper">
        <div class="container">
            <header>
                <table width="100%" align="center" cellpadding="0" cellspacing="2">
                <tr>
                <th width="20%"><img src="jblogo.jpg" alt="JBS LOGO"  width="130"
      height="100"/></th>
                <th colspan=4>
                <h1 style="color:white;">JBS - WORLD BEST ONLINE EBOOKS WEBSITE</h1>
                </th>
                </tr>
                </table>
            </header>
            <nav>
                <table width="100%" align="center" cellpadding="0" cellspacing="2">
                <tbody align="center"  style="font-weight:bold;font-size:18px;"">
                <tr>
                <td width="20%"><hr><a href="index.html">Home</a><hr></td>
                <td width="20%"><hr><a href="login.html">Login</a><hr></td>
                <td width="20%"><hr><a href="registration.html">Registration</a><hr></td>
                <td width="20%"><hr><a href="cart.html" >Cart</a><hr></td>
                </tr>
                </tbody>
                </table>
            </nav>
        </div>
        <div class="container1">
            <div class="sidebar1"></div>
            <div class="container2">
                <main>
                    <center><br>
                        <h3>Registration Form </h3>
                        <br/>
                        <form name="f1">
                        <table cellpadding="1" align="center" >
                        <tr><td> Name:*</td>
                        <td><input type="text" name="username"></td></tr>
                        <tr><td>Password:*</td>
                        <td><input type="password" name="password"></td></tr>
                        <tr><td>Email ID:*</td>
                        <td><input type="text" name="email"></td></tr>
                        <tr><td>Phone Number:*</td>
                        <td><input type="text" name="phno"></td></tr>
                        <tr><td valign="top">Gender:*</td>
```

```
                <td><input type="radio" name="radio" value="1">Male   
                <input type="radio" name="radio" value="2">Female</td></tr>
                <tr> <td valign="top">Language Known:*</td>
                <td> <input type="checkbox" name="checkbox" value="English">English<br/>
                <input type="checkbox" name="checkbox" value="Telugu">Telugu<br>
                <input type="checkbox" name="checkbox" value="Hindi">Hindi<br>
                <input type="checkbox" name="checkbox" value="Tamil">Tamil
                </td></tr>
                <tr> <td valign="top">Address:*</td>
                <td><textarea name="address"></textarea></td>
                <tr><td></td>
            <td><input type="submit" value="submit" hspace="10"  onclick="validate()">
                <input type="reset" value="reset"></td></tr>
                <tr> <td colspan=2 >*<font color="#FF0000">fields are mandatory</font>
                </td>
                </tr>
                </table>
                </form>
              </center>
            </main>
          </div>
          <div class="sidebar2"></div>
        </div>
        <footer><font color="white">(C) 2024 All rights reserved by JBS ebooks</font></footer>
      </div>
    </body>
    </html>
```

Index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Home - JB Book Store</title>
<!-- Bootstrap CSS -->
<link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0-
alpha1/dist/css/bootstrap.min.css" rel="stylesheet">
<style>
body {
background-color: #f8f9fa;
}
.header-img {
height: 100px;
}
.custom-header {
background-color: #ffffff;
color: white;
padding: 20px 0;
}
.custom-nav a {
color: white;
text-decoration: none;
```

```
padding: 10px 20px;
display: inline-block;
}
.custom-nav a:hover {
border-radius: 5px;
}
nav{
background-color:#0b6620;
}
</style>
<link
href="https://maxcdn.bootstrapcdn.com/bootstrap/4.5.2/css/bootstrap.min.css"
rel="stylesheet">
</head>
<body>
<div class="container-fluid">
<div class="container-fluid">
<!-- Header Section -->
<header class="custom-header text-center">
<div class="container d-flex align-items-center justify-contentbetween">
<img src="jblogo.jpg" alt="JB Logo" class="header-img">
<h1 class="mb-0">JB Online Book Store - Best Online Ebooks Store</h1>
</div>
</header>
<!-- Navigation Section -->
<nav class="custom-nav text-center">
<a href="index.html">Home</a>
<a href="login.html">Login</a>
<a href="registration.html">Register</a>
<a href="cart.html">Cart</a>
</nav>
<!-- Main Content -->
<main class="mt-4">
<div class="container text-center">
<h2>Welcome to JB Online Book Store </h2>
<img src="jblogo.jpg" alt="" width="10%" height="10%">
<p>Your one-stop destination for online eBooks.</p>
<button class="btn btn-primary">Explore Now</button>
</div>
</main>
<!-- Footer Section -->
<footer class="bg-dark text-white text-center py-3 mt-4">
<p>&copy; 2026 JB Online Book Store. All rights reserved.</p>
</footer>
</div>
<!-- Bootstrap JS -->
<script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0-
alpha1/dist/js/bootstrap.bundle.min.js"></script>
</div>
</body>
</html>
```

Login.html

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Login - JB Book Store</title>
<!-- Bootstrap CSS -->
<link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0-
alpha1/dist/css/bootstrap.min.css" rel="stylesheet">
<style>
body {
background-color: #f8f9fa;
}
.header-img {
height: 100px;
}
.custom-header {
background-color: #ffffff;
color: white;
padding: 20px 0;
}
.custom-nav a {
color: white;
text-decoration: none;
padding: 10px 20px;
display: inline-block;
}
.custom-nav a:hover {
background-color: #495057;
border-radius: 5px;
}
</style>
<link
href="https://maxcdn.bootstrapcdn.com/bootstrap/4.5.2/css/bootstrap.min.css"
rel="stylesheet">
</head>
<body>
<div class="container-fluid">
<div class="container-fluid">
<!-- Header Section -->
<header class="custom-header text-center">
<div class="container d-flex align-items-center justify-contentbetween">
<img src="jblogo.jpg" alt="JB LOGO" class="header-img">
<h1 class="mb-0">JB Online Book Store - Best Online Ebooks Store</h1>
</div>
</header>
<!-- Navigation Section -->
<nav class="bg-dark custom-nav text-center">
<a href="index.html">Home</a>
<a href="login.html">Login</a>
<a href="registration.html">Register</a>
<a href="cart.html">Cart</a>
</nav>
<!-- Login Form Section -->
```

```
<main class="mt-4">
<div class="container">
<h2 class="text-center">Login</h2>
<form class="mx-auto mt-3" style="max-width: 400px;">
<div class="mb-3">
<label for="username" class="form-label">Username</label>
<input type="text" class="form-control" id="username"
placeholder="Enter your username">
</div>
<div class="mb-3">
<label for="password" class="form-label">Password</label>
<input type="password" class="form-control" id="password"
placeholder="Enter your password">
</div>
<div class="mb-3 form-check">
<input type="checkbox" class="form-check-input"
id="rememberMe">
<label class="form-check-label" for="rememberMe">Remember
Me</label>
</div>
<button type="submit" class="btn btn-primary w-
100">Login</button>
</form>
</div>
</main>
<!-- Footer Section -->
<footer class="bg-dark text-white text-center py-3 mt-4">
<p>&copy; 2026 JB Online Book Store. All rights reserved.</p>
</footer>
</div>
<!-- Bootstrap JS -->
<script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0-
alpha1/dist/js/bootstrap.bundle.min.js"></script>
</div>
</body>
</html>
```

Registration.html

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Registration Page</title>
<!-- Bootstrap CSS -->
<link
href="https://cdnjs.cloudflare.com/ajax/libs/bootstrap/5.3.0/css/bootstrap.min.cs
s" rel="stylesheet">
<link rel="stylesheet" href="stylecss">
<style>
body {
background-color: #f8f9fa;
}
```

```
.header-img {
height: 100px;
}
.custom-header {
background-color: #ffffff;
color: white;
padding: 20px 0;
}
.custom-nav a {
color: white;
text-decoration: none;
padding: 10px 20px;
display: inline-block;
}
.custom-nav a:hover {
background-color: #495057;
border-radius: 5px;
}
</style>
<link
href="https://maxcdn.bootstrapcdn.com/bootstrap/4.5.2/css/bootstrap.min.css"
rel="stylesheet">
</head>
<body>
<div class="container-fluid">
<!-- Header Section -->
<header class="custom-header text-center">
<div class="container d-flex align-items-center justify-content-between">
<img src="jblogo.jpg" alt="jb logo" class="header-img">
<h1 class="mb-0">JB Online Book Store - Best Online Ebooks Store</h1>
</div>
</header>
<!-- Navigation Section -->
<nav class="bg-dark custom-nav text-center">
<a href="index.html">Home</a>
<a href="login.html">Login</a>
<a href="registration.html">Register</a>
<a href="cart.html">Cart</a>
</nav>
<!-- Main Content -->
<main class="container my-5">
<h1 class="text-center mb-4">Registration</h1>
<form class="row g-3">
<div class="col-md-6">
<label for="firstName" class="form-label">First Name</label>
<input type="text" class="form-control" id="firstName"
placeholder="Enter your first name">
</div>
<div class="col-md-6">
<label for="lastName" class="form-label">Last Name</label>
<input type="text" class="form-control" id="lastName"
placeholder="Enter your last name">
</div>
<div class="col-md-12">
```

```
<label for="email" class="form-label">Email</label>
<input type="email" class="form-control" id="email"
placeholder="Enter your email">
</div>
<div class="col-md-12">
<label for="password" class="form-label">Password</label>
<input type="password" class="form-control" id="password"
placeholder="Enter your password">
</div>
<div class="col-md-12">
<label for="confirmPassword" class="form-label">Confirm
Password</label>
<input type="password" class="form-control" id="confirmPassword"
placeholder="Confirm your password">
</div>
<div class="col-12">
<button type="submit" class="btn btn-primary w-
100">Register</button>
</div>
</form>
</main>
<!-- Footer -->
<footer class="bg-dark text-white text-center py-3">
<p class="mb-0">&copy; 2026 JB Online Book Store. All rights reserved.</p>
</footer>
<!-- Bootstrap JS -->
<script
src="https://cdnjs.cloudflare.com/ajax/libs/bootstrap/5.3.0/js/bootstrap.bundle.m
in.js"></script>
</div>
</body>
</html>
```

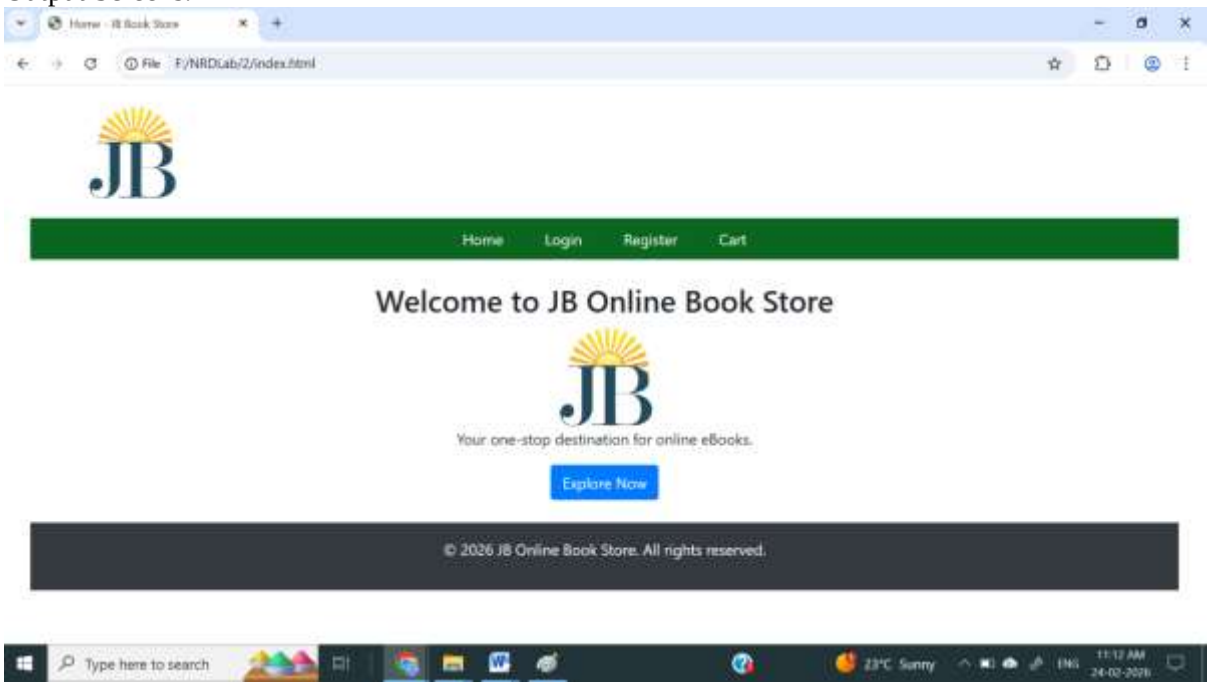
Cart.html

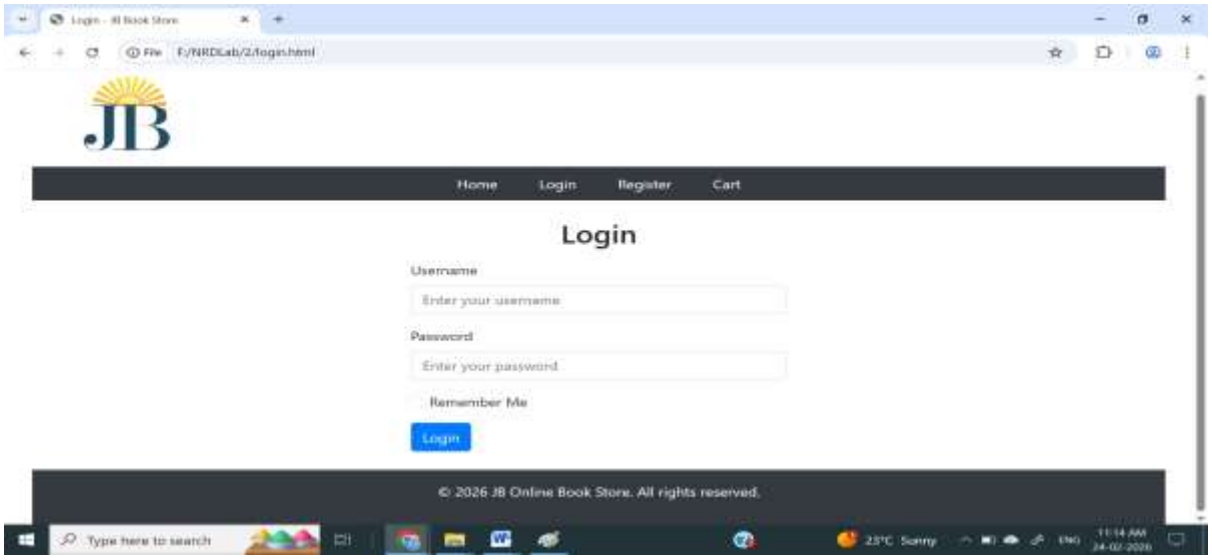
```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Cart Page</title>
<!-- Bootstrap CSS -->
<link
href="https://cdnjs.cloudflare.com/ajax/libs/bootstrap/5.3.0/css/bootstrap.min.cs
s" rel="stylesheet">
<link rel="stylesheet" href="stylecss">
<style>
body {
background-color: #f8f9fa;
}
.header-img {
height: 100px;
}
.custom-header {
```

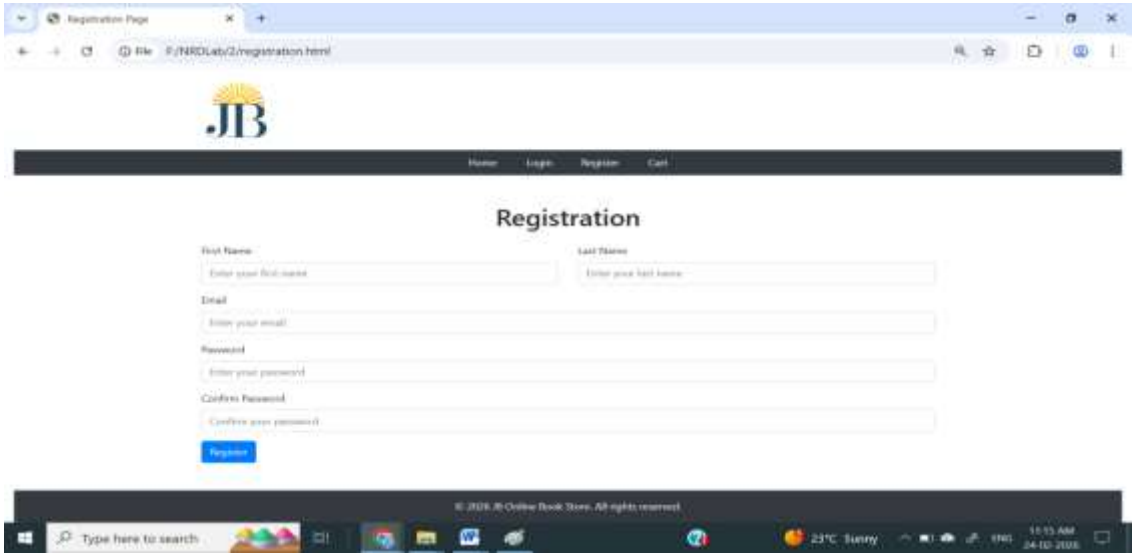
```
background-color: #ffffff;
color: white;
padding: 20px 0;
}
.custom-nav a {
color: white;
text-decoration: none;
padding: 10px 20px;
display: inline-block;
}
.custom-nav a:hover {
background-color: #495057;
border-radius: 5px;
}
</style>
<link
href="https://maxcdn.bootstrapcdn.com/bootstrap/4.5.2/css/bootstrap.min.css"
rel="stylesheet">
</head>
<body>
<div class="container-fluid">
<!-- Header Section -->
<header class="custom-header text-center">
<div class="container d-flex align-items-center justify-content-between">
<img src="jblogo.jpg" alt="JB LOGO" class="header-img">
<h1 class="mb-0">JB Online Book Store - Best Online Ebooks Store</h1>
</div>
</header>
<!-- Navigation Section -->
<nav class="bg-dark custom-nav text-center">
<a href="index.html">Home</a>
<a href="login.html">Login</a>
<a href="registration.html">Register</a>
<a href="cart.html">Cart</a>
</nav>
<!-- Main Content -->
<main class="container my-5">
<h1 class="text-center mb-4">Your Cart</h1>
<div class="table-responsive">
<table class="table table-bordered">
<thead class="table-dark">
<tr>
<th>Item</th>
<th>Description</th>
<th>Price</th>
<th>Quantity</th>
<th>Total</th>
<th>Actions</th>
</tr>
</thead>
<tbody>
<tr>
<td> 1 </td>
<td>Java Programming</td>
```

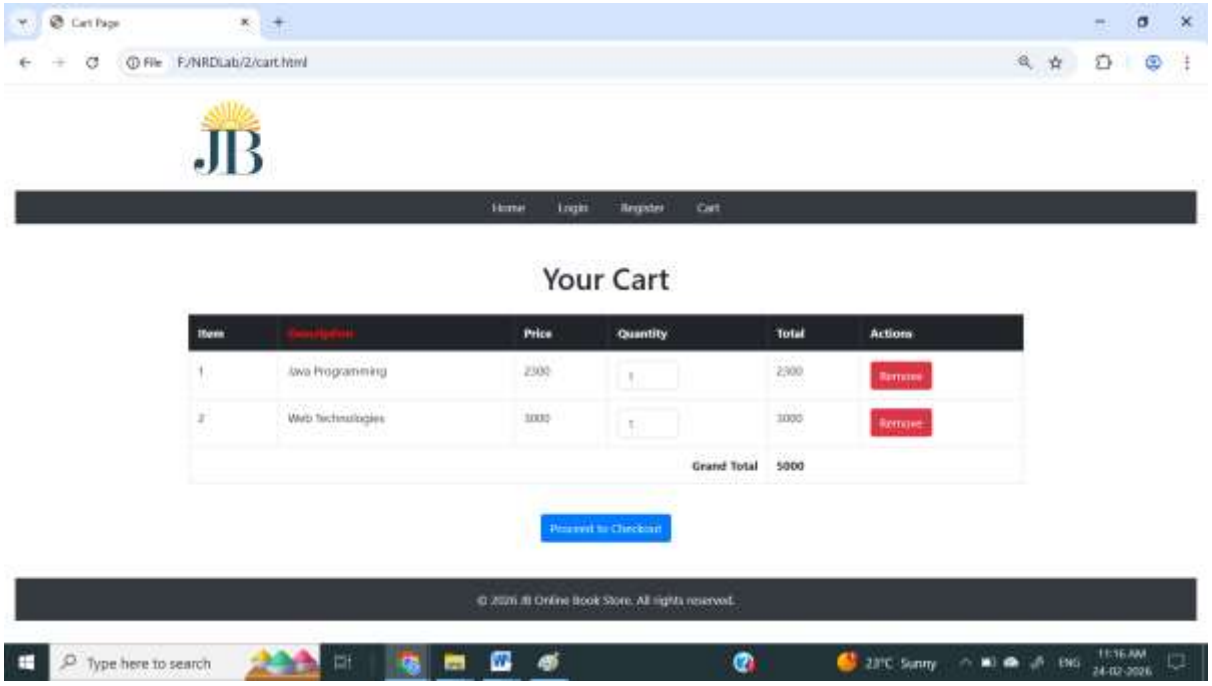
```
<td> 2300</td>
<td><input type="number" value="1" class="form-control"
style="width: 80px;"></td>
<td>2300</td>
<td><button class="btn btn-danger btnsm">
Remove</button></td>
</tr>
<tr>
<td>2 </td>
<td>Web Technologies</td>
<td>3000</td>
<td><input type="number" value="1" class="form-control"
style="width: 80px;"></td>
<td>3000</td>
<td><button class="btn btn-danger btnsm">
Remove</button></td>
</tr>
</tbody>
<tfoot>
<tr>
<th colspan="4" class="text-end">Grand Total</th>
<th colspan="2">5000</th>
</tr>
</tfoot>
</table>
</div>
<div class="text-center mt-4">
<button class="btn btn-primary">Proceed to Checkout</button>
</div>
</main>
<!-- Footer -->
<footer class="bg-dark text-white text-center py-3">
<p class="mb-0">&copy; 2026 JB Online Book Store. All rights reserved.</p>
</footer>
<!-- Bootstrap JS -->
<script
src="https://cdnjs.cloudflare.com/ajax/libs/bootstrap/5.3.0/js/bootstrap.bundle.m
in.js"></script>
</div>
</body>
</html>
```

Output Screens:








Cart Page
F:/NRDLab/2/cart.html
JB
Home
Login
Register
Cart
Your Cart
Item
Description
Price
Quantity
Total
Actions
1
Java Programming
2300
1
2300
Remove
2
Web Technologies
3000
1
3000
Remove
Grand Total
5000
Proceed to Checkout
© 2026 JB Online Book Store. All rights reserved.
Type here to search
23°C Sunny
ENG
11:16 AM
24-02-2026

3. Use JavaScript for doing client – side validation of the pages implemented in experiment 1 and experiment 2.

Client - Side validation of Registration Page

registrationJS.html

```
<html>
<head>
   <title> Welcome to NNRG e-Book's website</title>
   <script language="javascript">
   function validate() {
      // username validation
      var uname = f1.username.value;
      if (uname.length<=0)
      {
         alert("Please Enter UserName");
         f1.username.focus();
         return false;
      }
      if (uname.length < 8)
      {
         alert("Please enter UserName not less than 8");
         f1.username.focus();
         return false;
      }
      //password validation
      var pwd = f1.password.value;
      if (pwd.length<=0)
      {
         alert("Please Enter password");
         f1.password.focus();
         return false;
      }
      if (pwd.length < 6)
      {
         alert("Please enter Password not less than 6");
         f1.password.focus();
         return false;
      }
      // email validation
      var email = f1.email.value;
      if (email.length<=0)
      {
         alert("Please Enter email");
         f1.email.focus();
         return false;
      }
      else {
         let eflag=false;
         for(i=0;i<email.length;i++) {
            if(email.charAt(i)=="@")
               {
                  eflag=true;
```

```
            }
        }
        if(!(eflag))
        {
            alert("Please enter a valid Email ID");
            f1.email.focus();
            return false;
        }
    }
    // phone number validation
    var phno = f1.phno.value;
    if (phno.length<=0)
    {
        alert("Please Enter Phone Number");
        f1.phno.focus();
        return false;
    }
    if (isNaN(phno))
    {
        alert("Please Enter Valid Phone Number");
        f1.phno.focus();
        return false;
    }
    if (phno.length != 10)
    {
        alert("Please Enter Valid Phone Number");
        f1.phno.focus();
        return false;
    }
    // gender validation
    let flag=false;
    for(i=0;i<f1.gen.length;i++)
        if(f1.gen[i].checked)
            flag=true;
    if(!(flag))
    {
    alert("Please choose a Gender");
    return false;
    }
    // Language validation
    flag=false;
    for(i=0;i<f1.lang.length;i++)
        if(f1.lang[i].checked)
            flag=true;
    if(!(flag))
    {
    alert("Please select at least one of the Language options.");
    return false;
    }
    // address validation
    var addr = f1.address.value;
    if (addr.length<=0)
    {
        alert("Please Enter address");
```

```
        f1.address.focus();
        return false;
      }
      // to display Success message
      alert("Registration Successful");
    }
    </script>
  </head>
  <body>
  <center><br>
    <h3>Registration Form </h3>
    <br/>
    <form name="f1">
    <table cellpadding="1" align="center" >
    <tr><td> User Name:*</td>
    <td><input type="text" name="username"></td></tr>
    <tr><td>Password:*</td>
    <td><input type="password" name="password"></td></tr>
    <tr><td>Email ID:*</td>
    <td><input type="text" name="email"></td></tr>
    <tr><td>Phone Number:*</td>
    <td><input type="text" name="phno"></td></tr>
    <tr><td valign="top">Gender:*</td>
    <td><input type="radio" name="gen" value="Male">Male   
    <input type="radio" name="gen" value="Female">Female</td></tr>
    <tr> <td valign="top">Language Known:*</td>
    <td> <input type="checkbox" name="lang" value="English">English<br/>
    <input type="checkbox" name="lang" value="Telugu">Telugu<br>
    <input type="checkbox" name="lang" value="Hindi">Hindi<br>
    <input type="checkbox" name="lang" value="Tamil">Tamil
    </td></tr>
    <tr> <td valign="top">Address:*</td>
    <td><textarea name="address"></textarea></td>
    <tr><td></td><td><input type="button" value="SUBMIT" hspace="10"
      onclick="validate()">
    <input type="reset" value="RESET"></td></tr>
    <tr> <td colspan=2 >*<font color="#FF0000">fields are mandatory</font>
    </td>
    </tr>
    </table>
    </form>
  </center>
  </body>
  </html>
```

i. **Add onsubmit="return validateLogin()" to your <form> tag of Login.html**
   Syntax
   <form name="f1" onsubmit="return validateLogin()">

ii. **Add a <script> tag just before your closing </body> tag and paste the function inside it:**

```
<script>
 function validateLogin() {
const username = document.f1.username.value.trim();
const password = document.f1.password.value.trim();
```

```
  // Username validation
  if (username === "") {
    alert("Username cannot be empty.");
    document.f1.username.focus();
    return false;
  }

  if (username.length < 4) {
    alert("Username must be at least 4 characters long.");
    document.f1.username.focus();
    return false;
  }

  // Password validation
  if (password === "") {
    alert("Password cannot be empty.");
    document.f1.password.focus();
    return false;
  }
  if (password.length < 6) {
    alert("Password must be at least 6 characters long.");
    document.f1.password.focus();
    return false;
  }
  const hasUpperCase = /[A-Z]/.test(password);
  const hasLowerCase = /[a-z]/.test(password);
  const hasNumber   = /[0-9]/.test(password);

  if (!hasUpperCase || !hasLowerCase || !hasNumber) {
    alert("Password must contain at least one uppercase letter, one lowercase letter, and one
number.");
    document.f1.password.focus();
    return false;
  }

  alert("Login successful! Welcome, " + username + "!");
  return true;
}
  </script>
```

4. **Explore the features of ES6 like arrow functions, callbacks, promises, async/await. Implement an application for reading the weather information from openweathermap.org and display the information in the form of a graph on the web page.**

Index.html

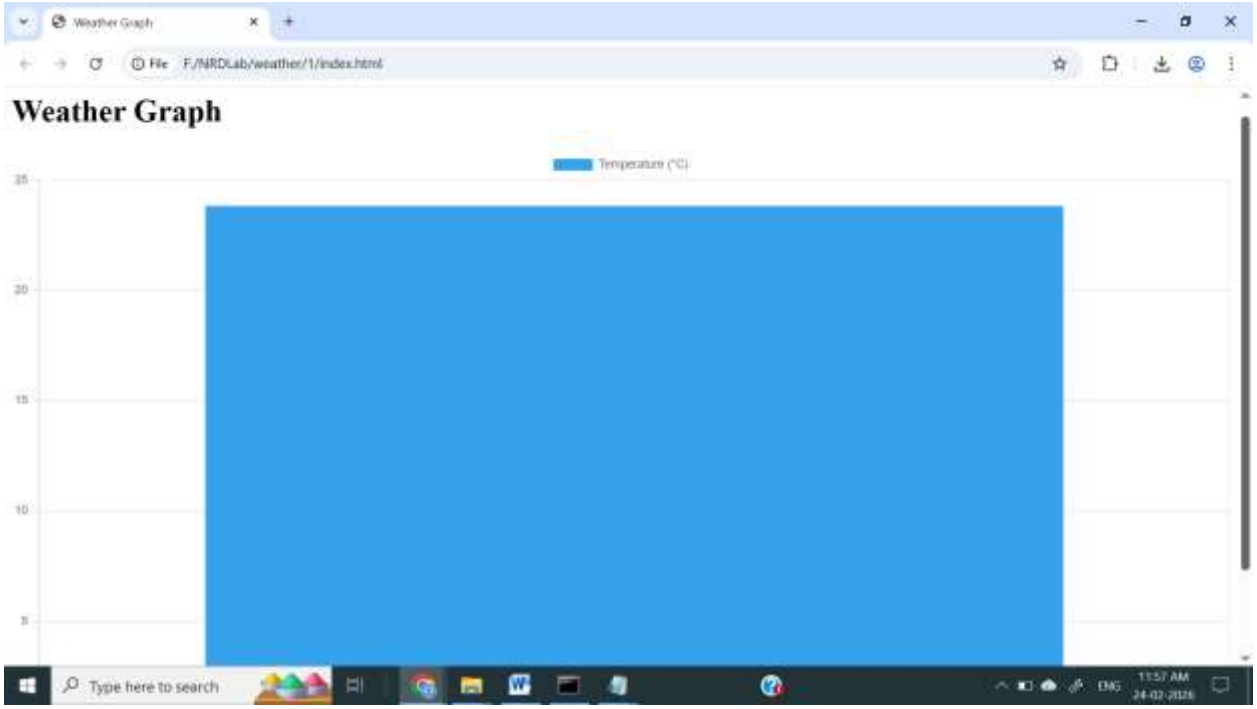
```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <link rel="stylesheet" href="styles.css">
  <title>Weather Graph</title>
</head>
<body>
  <div class="container">
    <h1>Weather Graph</h1>
    <canvas id="weatherGraph" width="400" height="200"></canvas>
  </div>
  <!-- Include third-party libraries from their CDNs -->
  <script src="https://cdn.jsdelivr.net/npm/axios/dist/axios.min.js"></script>
  <script src="https://cdn.jsdelivr.net/npm/chart.js"></script>
  <!-- Link to your local script file -->
  <script src="script.js"></script>
</body>
</html>
```

script.js

```
document.addEventListener('DOMContentLoaded', () => {
  const apiKey = 'e4487e40ea5b6d108d6382f7bd465cd6';
  const city = 'Hyderabad';
  const apiUrl =
`https://api.openweathermap.org/data/2.5/weather?q=${city}&appid=${apiKey}&units=m
etric`;

  // Fetch weather data from API
  const fetchData = async () => {
    try {
      const response = await axios.get(apiUrl);
      const weatherData = response.data;
      updateGraph(weatherData.main.temp);
    } catch (error) {
      console.error('Error fetching weather data:', error.message);
    }
  };

  // Update the chart with fetched temperature
  const updateGraph = (temperature) => {
    const ctx = document.getElementById('weatherGraph').getContext('2d');

    // Initialize Chart.js
```

```
        new Chart(ctx, {
          type: 'bar',
          data: {
            labels: ['Temperature'],
            datasets: [{
              label: 'Temperature (°C)',
              data: [temperature],
              backgroundColor: ['#36A2EB'],
            }],
          },
          options: {
            scales: {
              y: {
                beginAtZero: true, // Fixed spacing error here
              },
            },
          },
        });
    };

    // Run on load
    fetchData();
});
```

style.css

```
/* Styles the entire body of the HTML document */
body {
    font-family: 'Arial', sans-serif; /* Sets the primary font */
    margin: 0;                        /* Removes default browser margin */
    padding: 0;                       /* Removes default browser padding */
    background-color: #f4f4f4;        /* Sets a light gray background color */
}

/* Styles elements with the class 'container' (e.g., <div class="container">) */
.container {
    max-width: 600px;                 /* Limits the maximum width of the container */
    margin: 50px auto;                /* Centers the container horizontally and adds top/bottom
margin */
    background-color: #fff;           /* Sets a white background */
    padding: 20px;                    /* Adds internal spacing around the content */
    border-radius: 8px;               /* Rounds the corners of the container */
    box-shadow: 0 0 10px rgba(0, 0, 0, 0.1); /* Adds a subtle shadow for depth */
}

/* Styles all <h1> heading elements */
h1 {
    text-align: center;               /* Centers the text within the heading */
}

/* Styles all <canvas> elements (typically used for graphics) */
canvas {
    display: block;                   /* Ensures the canvas takes up its own block of space */
```

```
    margin: 20px auto;              /* Centers the canvas horizontally and adds top/bottom
margin */
}
```